# Evaluation Report of EliteFS-Day23-AI-KMIT

Average Score: 7/10 | Attempted: 3 of 3 questions

## Question 1

THEORY REMEMBER Best Score: 8/10

What is the role of gamma and beta in Batch Normalization?

### Desired Answer

Gamma (γ) and Beta (β) are learnable parameters in Batch Normalization that allow the network to adjust the normalized output. After normalizing inputs to have zero mean and unit variance, γ scales the values and β shifts them. This ensures that the model can restore or modify the distribution if needed. Without γ and β, the normalization might limit the representational power of the layer.

### Student's Answer

During training the each layers distribution changes, therefore to ensure stability, batch normalization is used, which ensures for every batch the mean is 0 and variance is 1, But sometimes when the output is not mean=0, or variance=1, then it is not appropriate. therefore gamma, beta is used, y= gamma*(x)+ beta, gamma controls variance, beta controls mean.

### Evaluations

#### Feedback 1

★★★★☆ 8/10

The student correctly recalls that γ and β perform scaling and shifting (y = γ·x + β) after normalization, and that γ controls variance while β controls the mean. They also understand this is applied after ensuring zero-mean, unit-variance normalization. They did not explicitly state that γ and β are learnable parameters or mention the effect on representational power when they are omitted, which are secondary details in the Actual Answer. Hence, this is good recall of the primary role of γ and β but misses some minor points.

#### Feedback 2

★★★★☆ 8/10

The student correctly describes that batch normalization first enforces zero mean and unit

variance and then applies the transform $y = \gamma \cdot x + \beta$. They accurately state that $\gamma$ scales (controls variance) and $\beta$ shifts (controls mean), matching the core of the Actual Answer. They also imply that this adjustment is needed when strict normalization is not appropriate. They did not explicitly call $\gamma$ and $\beta$ 'learnable parameters' or mention representational power, which are minor omissions, so the response earns a good-recall score.

## Question 2

THEORY ANALYZE Best Score: 9/10

What is the difference between an eigenvector and an eigenvalue?

### Desired Answer

An eigenvector is a special vector whose direction does not change when a matrix transformation is applied to it. An eigenvalue is the scalar that tells how much that eigenvector is stretched or compressed during the transformation. In the equation $Av=\lambda v$, $v$ is the eigenvector and $\lambda$ is the eigenvalue. In simple terms, the eigenvector gives the direction, and the eigenvalue gives the scaling factor in that direction. In simple terms: Eigenvector = special direction, Eigenvalue = amount of scaling in that direction

### Student's Answer

Eigen Vector is a vector that does not change the direction when transformation is applied eigen value: how much transformation increased or decreased.

### Evaluations

**Feedback 1** ★★★★½ 9/10

The student is demonstrating understanding of the same concepts as the Actual Answer. They correctly state that an eigenvector is a vector whose direction remains unchanged by the transformation and that an eigenvalue quantifies how much that vector is stretched or compressed (increased or decreased). This matches the core ideas of direction versus scaling from the Actual Answer. The response is accurate and in the student's own words, though it is brief and omits explicitly calling the eigenvalue a scalar or showing the equation $Av=\lambda v$. These are minor details, so a score of 9/10 is appropriate.

### Feedback 2

★★★★☆ 8/10

The student is demonstrating understanding of the same concepts as the Actual Answer. They correctly identify that an eigenvector is a vector whose direction remains unchanged under a transformation and that an eigenvalue measures how much the transformation increases or decreases (i.e. stretches or compresses) that vector. These capture the two core ideas from the Actual Answer. The response is accurate and concise but omits the formal notation (Av = λv) and the explicit reference to the eigenvalue as a scalar, so there is only a minor omission in detail.

## Question 3

THEORY UNDERSTAND Best Score: 5/10

Why should the weights in Xavier initialization be in the range –sqrt(6/(nin+nout)) to + sqrt(6/(nin+nout))?

### Desired Answer

Xavier initialization chooses the range –sqrt(6/(nin+nout)) to + sqrt(6/(nin+nout)), so that the variance of the weights becomes 2/(nin+nout). This keeps the variance of activations and gradients approximately constant across layers. As a result, it prevents vanishing and exploding gradients during training. The specific range comes from matching the variance of a uniform distribution to this desired value.

### Student's Answer

As it prevents vanishing and exploding gradient and ensures uniform distribution

### Evaluations

#### Feedback 1

★½☆☆☆ 3/10

The student is addressing the same topic—Xavier initialization and its role in preventing vanishing/exploding gradients. However, they give only a very high-level answer: they note that the chosen range prevents gradient issues and that the distribution is uniform, but they

do not explain why that particular ±√(6/(nin+nout)) bound is chosen (i.e., matching the variance of a uniform distribution to 2/(nin+nout)). They miss the core derivation and reasoning behind the specific range. This shows a weak understanding with significant gaps, so a score of 3 is appropriate.

### Feedback 2

★★½☆☆ 5/10

The student is addressing the same concepts as the Actual Answer—namely, that Xavier initialization prevents vanishing/exploding gradients and uses a uniform distribution—but their answer is very brief and omits the core mechanistic detail. They do not explain that the specific ±√(6/(nin+nout)) range is chosen to match the variance of a uniform distribution to the target variance 2/(nin+nout), which keeps activations and gradients roughly constant across layers. Thus they show a basic grasp of the goal but miss the key variance-matching rationale.